passing through the strait would pay toll. A Venetian merchant ship refused to stop and shell out the tax; Turkish cannon blasted the ship to bits, killing most of the crew. The captain was hauled out of the water and impaled on the shore where other ships could see him.[14]

The storm was coming. Constantine set his people, that winter, to collecting weapons, repairing the walls, gathering provisions. A Hungarian cannon maker named Urban came to the city and offered his services; he was too expensive for Constantine, so with regret the emperor turned him away. He went to Cutter of the Strait instead, where Mehmet promptly hired him.[15]

At the beginning of April, Mehmet began to move his forces into place. He established his own headquarters right across from the gate known as St. Romanus and established other camps all along the walls. Monstrelet says that he had 200,000 men; the Venetian eyewitness Nicolò Barbaro guessed 160,000, the Greek chronicler Chalkokondyles 400,000. Among them were at least 60,000 archers and 40,000 horsemen; some of the foot soldiers armed only with scimitars, but others with iron helmets and French chain mail. "There were very many bombards and culverins," Monstrelet says; *bombards* were wide-mouthed cannon that hurled granite balls, *culverins* smaller cannon that could be fired by hand. One of the bombards, according to Monstrelet, shot out stones "weighing eighteen hundred pounds." This was the work of the renegade Hungarian cannon maker; it had taken sixty oxen pulling thirty wagons to haul it from its forging place in Edirne (once called Adrianople) to the city's walls, and road workers had spent two months fortifying bridges and roadways ahead of it; the stones it hurled were more than seven feet around. The sultan hoped to use these to break through the triple land walls west of the city. No attacking force had ever before succeeded in this, but no attacking force had ever used cannon that hurled seven-foot stones either.[16]

The bombardment began on April 4, 1453.* It continued for fifty-five days; on each one of those days, the cannon launched 100 to 120 massive stones against the city's walls. Barbaro, who was inside the city during the attack, records that the first Turkish attempt to storm through the gaps knocked in the walls came on April 18. It was driven back by the defenders, stones piled back into the walls.[17]

The most vulnerable wall of the city was the one lining the edge of the Golden Horn, the finger of water that ran from the Bosphorus inland. The Horn itself was blocked with a massive chain, the chain guarded by towers. But Mehmet had planned for this. His engineers had outfitted about seventy of his ships with wheels. In the night of April 22, his men hauled the ships by land over the hills north of the Golden Horn and then slipped them into the